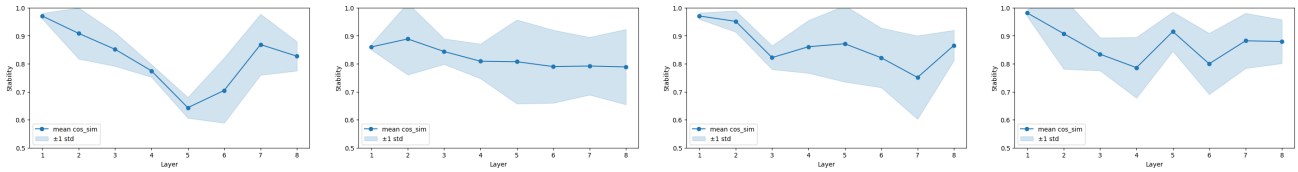
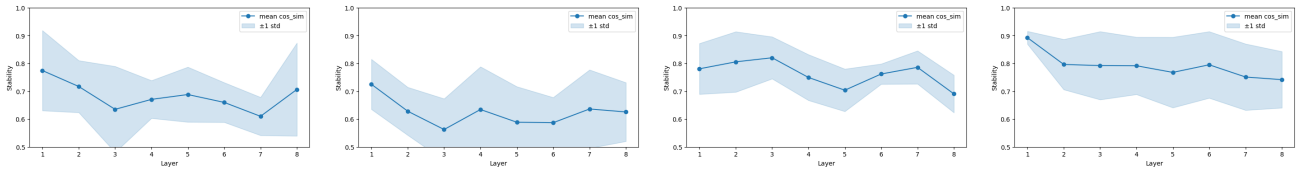
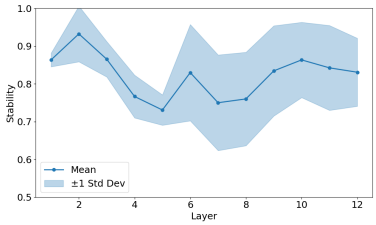


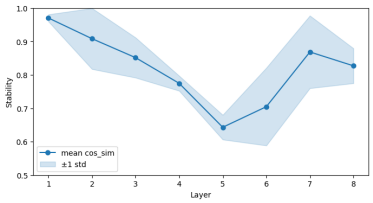
(a) Layers: 8, Heads: 8, Attn-only: False, Adam *(b)* Layers: 8, Heads: 16, Attn-only: False, Adam *(c)* Layers: 8, Heads: 8, Attn-only: False, AdamW *(d)* Layers: 8, Heads: 16, Attn-only: False, AdamW



(a) Layers: 8, Heads: 8, Attn-only: True, Adam *(b)* Layers: 8, Heads: 16, Attn-only: True, Adam *(c)* Layers: 8, Heads: 8, Attn-only: True, AdamW *(d)* Layers: 8, Heads: 16, Attn-only: True, AdamW



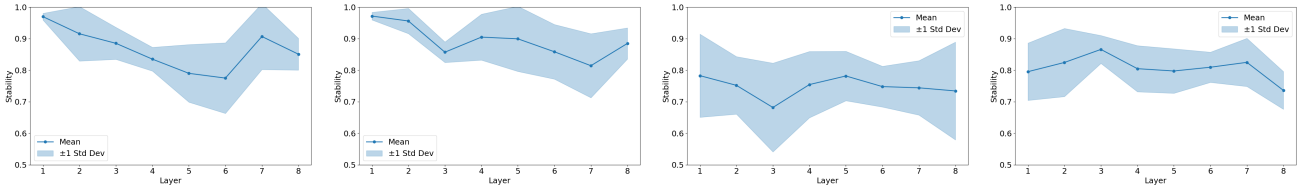
(a) Layers: 12, Heads: 12, Attn-only: False, Adam



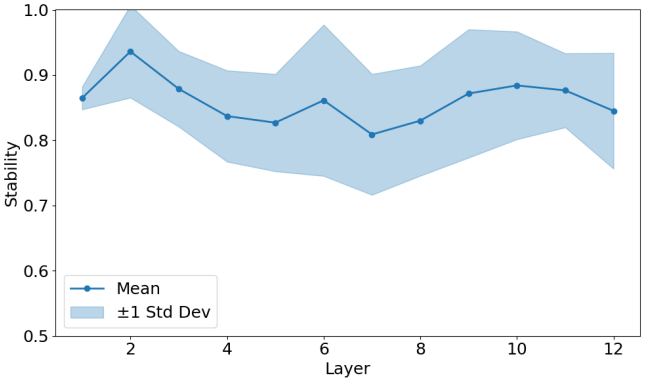
(b) Layers: 12, Heads: 12, Attn-only: False, AdamW

B.2. Cross-layer best-match stability

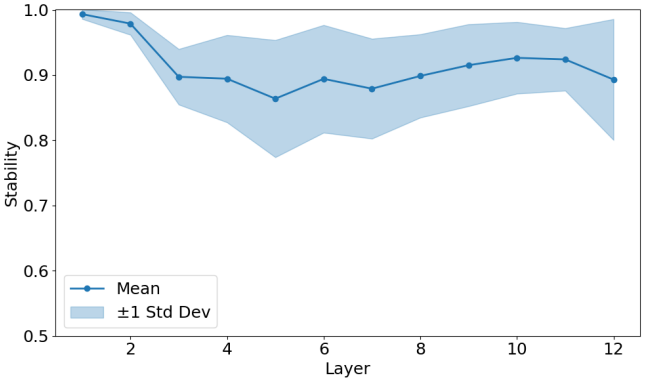
Representative plots for few of the architectures § 4.2:



(a) Layers: 8, Heads: 8, Attn-only: False, Adam *(b)* Layers: 8, Heads: 8, Attn-only: False, AdamW *(c)* Layers: 8, Heads: 8, Attn-only: True, Adam *(d)* Layers: 8, Heads: 8, Attn-only: True, AdamW

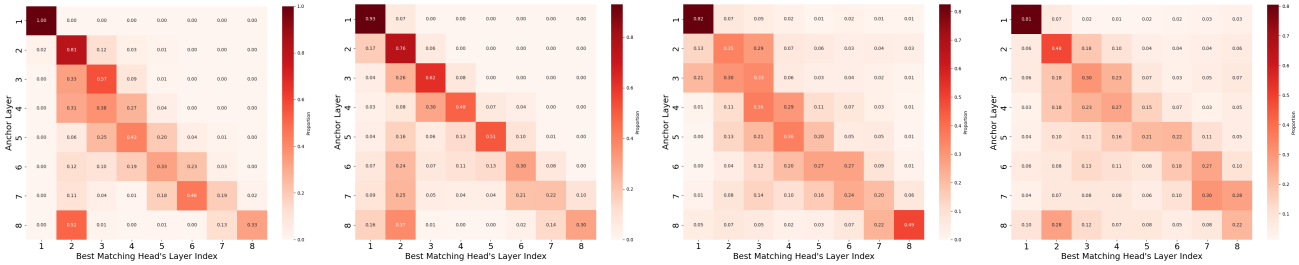


(e) Layers: 12, Heads: 12, Attn-only: False, Adam

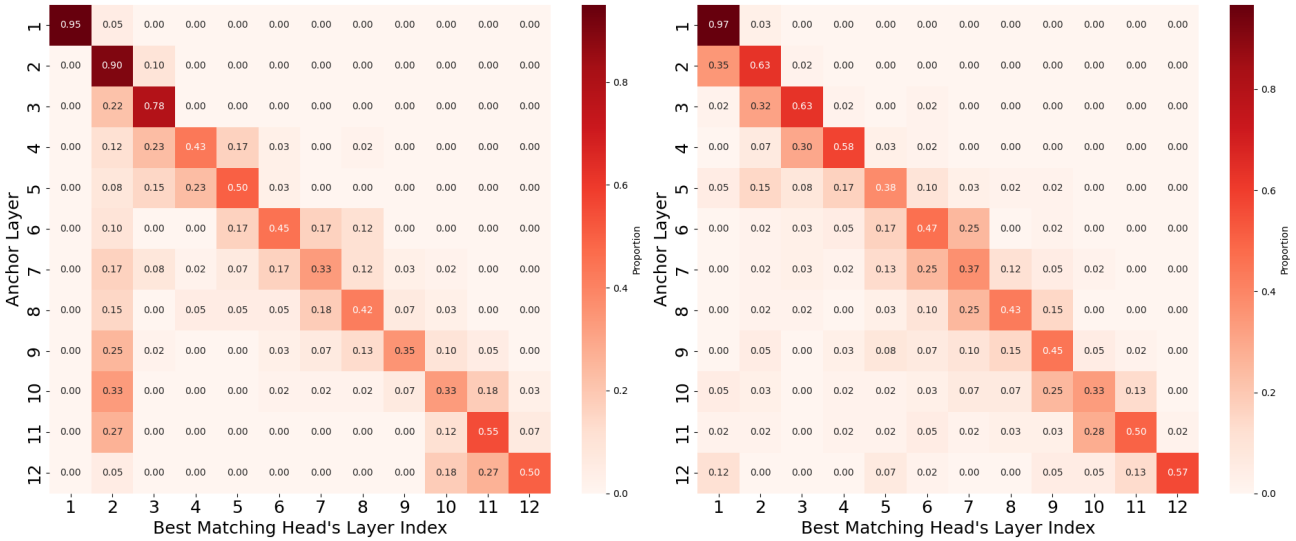


(f) Layers: 12, Heads: 12, Attn-only: False, AdamW

Figure 20. Representative plots for cross-layer best-match stability § 4.2



(a) Layers: 8, Heads: 8, Attn-only: False, Adam *(b)* Layers: 8, Heads: 8, Attn-only: False, AdamW *(c)* Layers: 8, Heads: 8, Attn-only: True, Adam *(d)* Layers: 8, Heads: 8, Attn-only: True, AdamW



(e) Layers: 12, Heads: 12, Attn-only: False, Adam *(f)* Layers: 12, Heads: 12, Attn-only: False, AdamW

Figure 21. Representative plots for alignment map & middle-layer dispersion § 4.2

B.3. Within-layer uniqueness of attention heads

Representative plots for few of the architectures § 4.3: